

Relationship

One to many:

- roles → users
One role can be assigned to many users.
- course_catalog → courses
One course definition (CSE101) can have many course offerings/sections.
- users → courses
One faculty member can teach many courses.
- users → course_files
One faculty member can have many course files.
- course_files → documents
One course file can contain multiple documents.
- required_items → documents
One required item type (item_no:1) can have many uploaded documents/files.
- users → refresh_tokens / one_time_tokens / audit_logs / import_batches
One user can have multiple refresh tokens, one-time tokens, audit logs, and import records.
- import_batches → courses
One PDF import process can create or update multiple courses.
- users → users (Self-relation)
One admin can create multiple chairpersons.

One-to-One:

- courses ↔ course_files - One course has exactly one course file.

Many-to-Many:

- courses ↔ required_items - One course can require multiple items, and the same item can be required by multiple courses.

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2. **Users**
3. **Refresh_tokens**
4. **One_time_tokens**
5. **Audit_logs** - Stores system activity history.
6. **Required_items** - Stores CQI required document (Manages the 1–17 required file)
7. **Course_catalog** - Stores course information. (course code, title, credit, course type, and default required items.)
8. **Courses** - course information with faculty, section, schedule, and semester.
9. **Course_files** - Represents the submission folder of a course.
10. **Documents** - Stores uploaded file metadata.
11. **Import_batches** - Tracks uploaded course PDFs, processing status, imported data, errors, and audit information.

Roles:

1. `_id` (ObjectId) - Primary identifier of the role document.
2. `name` (String) - Must be unique. Example: admin, chairperson, faculty.
3. `description` (String) - Description of the role purpose and responsibilities.
4. `is_system` (Boolean) - Indicates whether the role is a built-in system role. System roles are protected and cannot be deleted.
5. `created_at` (Date) - Stores the role creation timestamp.

Users:

1. `_id` (ObjectId) - Primary identifier of the user document.
2. `name` (String) - Full name of the user.
3. `email` (String) - Must be unique and stored in lowercase format. Used as the login identifier.
4. `password` (String | NULL) - Stores the encrypted password hash. (JWT token)
5. `auth_provider` (String) - Defines the authentication method used by the user.
6. Example values: local, google. Default value: local.
7. `google_id` (String | Null) - Stores the unique Google account ID provided by Google OAuth. Must be unique and sparse. Used for Google login authentication.
8. `profile_image` (Object | Null) - Stores the user's profile image information. Supports uploaded images and Google profile images.
9. `url` (String) - Stores the image URL/path.

10. `public_id` (String | Null) - Stores the cloud storage identifier (e.g., Cloudinary/S3 reference).
11. `provider` (String) - Defines the image source. Example values: `upload`, `google`.
12. `role` (String) - Defines the user role.
13. `short_code` (String) - Unique sparse field used as the faculty initials key.
Example: `RDA`. Used by the excel importer to resolve and link faculty information with imported courses.
14. `department` (String) - User's department. Default value: `CSE`.
15. `designation` (String) - Stores the user's academic/professional designation.
16. `employee_id` (String) - Stores the employee identification number.
17. `status` (String) - Defines the account status. Allowed values: `active`, `inactive`, `suspended`. Default value: `active`.
18. `is_email_verified` (Boolean) - Indicates whether the user's email has been verified. Default value: `false`.
19. `last_login_at` (Date) - Stores the user's last successful login timestamp.
20. `created_by` (ObjectId | Null) - Stores the user ID of the admin who created this account. `null` if the user registered themselves. Reference: `users._id`.
21. `created_at` (Date) - Stores the user account creation timestamp.
22. `updated_at` (Date) - Stores the last update timestamp of the user

Refresh_tokens:

1. `_id` (ObjectId) - Primary identifier of the refresh token document.
2. `user` (ObjectId) - Reference to the user who owns this refresh token.
3. `jti` (String) - Unique JWT identifier (UUID) embedded inside the refresh token. Must be unique.
4. `token_hash` (String) - Stores the SHA-256 hash of the raw refresh token. Raw tokens are never stored in the database for security reasons.
5. `expires_at` (Date) - Stores the refresh token expiration time. A TTL index automatically removes the document after expiry.
6. `revoked` (Boolean) - Indicates whether the refresh token has been invalidated. Default value: `false`.
7. `replaced_by` (String | Null) - Stores the `jti` of the new refresh token created after rotation. Used for refresh token reuse detection.
8. `user_agent` (String) - Stores the device/browser information from where the token was issued.
9. `ip` (String) - Stores the IP address from where the token was created. Used for security monitoring and suspicious activity detection.
10. `created_at` (Date) - Stores the refresh token creation timestamp.

One time token:

1. `_id` (ObjectId) - Primary identifier of the one-time token document.
2. `user` (ObjectId) - Reference to the user associated with this token.
3. `purpose` (String) - Defines the reason for generating the token.
Allowed values: `email_verify`, `password_reset`.
4. `token_hash` (String) - Stores the SHA-256 hash of the emailed token. The original raw token is never stored for security reasons.
5. `expires_at` (Date) - Stores the token expiration time. A TTL index automatically removes the token document after expiry.
6. `used_at` (Date | Null) - Stores the timestamp when the token was used.
If this field is not `null`, the token cannot be reused (single-use protection).
7. `created_at` (Date) - Stores the token creation timestamp.

Audit_Logs:

1. `_id` (ObjectId) - Primary identifier of the audit log document.
2. `actor` (ObjectId | Null) - Stores the user who performed the action.
`null` means the action was performed by the system or anonymous source.
Reference: `users._id`.
3. `action` (String) - Defines the performed action.
4. `target_type` (String | Null) - Defines the type of resource affected.
5. `target_id` (ObjectId | Null) - Stores the ID of the affected resource.
6. `meta` (Object) - Stores additional context information.
7. `ip` (String) - Stores the IP address from where the action was performed.
8. `created_at` (Date) - Stores the audit log creation timestamp.

Required_items:

1. `_id` (ObjectId) - Primary identifier of the required item document.
2. `item_no` (Int) - Unique item number.
3. `name` (String) - Name of the required document/item.
4. `description` (String) - Description of the item's purpose.
5. `category` (String) - Defines the item category.
Example values: `grades`, `obe`, `assessment`, `attendance`, `project`.
6. `has_sub_items` (Boolean) - Indicates whether this item contains sub-items.
Default value: `false`.
7. `sub_items` (Array<Object>) - Stores child items when required.
8. `allowed_extensions` (Array<String>) - Allowed file extensions for this item.
9. `max_size_mb` (Int) - Maximum allowed file size for this item.
10. `is_mandatory` (Boolean) - Indicates whether this item is required.
Default value: `true`.

11. active (Boolean) - Soft delete flag.
12. created_at (Date) - Stores creation timestamp.
13. updated_at (Date) - Stores last update timestamp.

Course_catalog:

1. _id (ObjectId) - Primary identifier of the course catalog document.
2. course_code (String) - Unique course identifier.
3. title (String) - Course title.
4. department (String) - Course department.
5. course_type (String) - Defines course category.
Allowed values: theory, lab, project, thesis, other.
6. credit_hours (Number) - Defines course credit value.
7. default_required_items (Array<Int>) - Stores default required document item numbers.
8. active (Boolean) - Defines whether the course is active.
Default value: true.
9. created_at (Date) - Stores creation timestamp.
10. updated_at (Date) - Stores last update timestamp.

Courses:

1. _id (ObjectId) - Primary identifier of the course offering document.
2. course_code (String) - Reference to course_catalog.course_code.
3. section (String) - Course section identifier.
4. title (String | Null) - Snapshot of course title during import.
5. semester (String) - Defines the semester.
6. course_type (String) - Course type from catalog.
7. department (String) - Derived from course code.
8. faculty (ObjectId | Null) - Resolved faculty reference.
9. faculty_code (String | Null) - Raw faculty initials from PDF.
10. capacity (Object)
11. enrolled (Int) - Number of enrolled students.
12. total (Int) - Maximum capacity.
13. schedule (Array<Object>) - Stores class schedule.
Fields:
 - day
 - start_time
 - end_time
 - room
14. source (Object) - Import information.
Fields:

- import_batch
 - file_name
 - imported_at
- 15.created_at (Date) - Creation timestamp.
- 16.updated_at (Date) - Update timestamp.

Course_Files:

1. _id (ObjectId) - Primary identifier of the course file document.
2. course (ObjectId) - Linked course reference.
3. faculty (ObjectId) - Course owner faculty.
4. semester (String) - Course semester.
5. status (String) - Workflow status.
Values: draft, submitted, under_review, approved, rejected.
6. review (Object)
 - reviewed_by (ObjectId)
 - comment (String)
 - reviewed_at (Date)
7. submitted_at (Date) - Submission timestamp.
8. created_at (Date) - Creation timestamp.
9. updated_at (Date) - Update timestamp.

Documents:

1. _id (ObjectId) - Primary identifier of uploaded document.
2. course_file (ObjectId) - Reference to course file.
3. course (ObjectId) - Denormalized course reference.
4. item_no (Int | Null) - Required item number.
5. sub_item (String | Null) - Sub-item key.
6. is_additional (Boolean) - Indicates extra faculty uploaded documents.
7. storage (Object)
 - original_name
 - file_name
 - file_path
 - mime_type
 - Size
8. processing (Object)
 - status(pending, completed)
9. review (Object)
 - status
- 10.uploaded_by (ObjectId) - User who uploaded the document.

- 11. created_at (Date) - Creation timestamp.
- 12. updated_at (Date) - Update timestamp.

Import_Batches:

- 1. _id (ObjectId) - Primary identifier of import batch.
- 2. kind (String) - Import type.
Example: offered_courses_pdf.
- 3. semester (String) - Semester extracted from PDF.
- 4. file_name (String) - Uploaded PDF filename.
- 5. file_hash (String) - SHA-256 hash of uploaded file.
- 6. department_filter (Array<String>) - Departments included in import.
- 7. status (String) - Import workflow status.
Values: preview, committed, failed.
- 8. errors (Array<Object>)
 - line
 - message
- 9. uploaded_by (ObjectId) - User who uploaded the file.
- 10. created_at (Date) - Creation timestamp.
- 11. committed_at (Date | Null) - Time when import was committed.